Hypercube Sums via a Dual Kronecker Chain: Sumcheck on Growing Domains, Polynomial-Valued FRI on Shrinking Domains

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Abstract

We study an interactive oracle proof for the Boolean-hypercube sum $H = \sum_{b\in\{0,1\}^m} f(b)$ of a committed multilinear polynomial f. The protocol runs two synchronized chains on the same challenges $r_1, \ldots, r_m$: a forward Kronecker–Sumcheck chain on growing multiplicative domains $D_0 \subset \cdots \subset D_m$, and a backward polynomial-valued (symbolic) FRI chain on shrinking domains, whose leaves are polynomials in a formal variable X. Both chains terminate at $f(r_1 X^{2^{m-1}}, \ldots, r_m X)$. We prove completeness for every m, show that in the exact-identity model both chains are deterministic bijections, give two explicit attacks on under-specified variants (one accepting a false sum with probability 1), and prove a soundness theorem for the repaired protocol with the explicit bound $m/|K| + 2B(2^m - 1)/|K| + (1-\theta)^q$, $\theta = (1 - 1/B)/2$. For Goldilocks^3, $m = 20$, $B = 4$, this gives $q = 189$ queries for 128-bit interactive soundness. The binding argument for the polynomial-valued chain is proximity-gap free: it uses only injectivity of the formal fold. Its price is a proof of size $\Theta(2^m)$. Specializing $X := 1$ collapses the protocol to classical sumcheck interleaved with scalar FRI (the BaseFold pattern); we isolate exactly which step then requires a proximity-gap theorem.

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1 Setting and notation

Fields. $k = \mathbb{F}_p$ with $p = 2^{64} - 2^{32} + 1$ (Goldilocks), and $K = \mathbb{F}_{p^3}$ a fixed cubic extension. All challenges are sampled uniformly from K.

Parameters. $m \geq 1$ is the number of variables, $L = 2^m$, $B \geq 2$ is the blowup (a power of two), $n = BL$. We require $n \mid p-1$. The rate is $\rho = 1/B$ and the unique-decoding threshold is

$$\theta := \frac{1-\rho}{2}.$$

Domains. For $0 \leq i \leq m$ let $D_i = \mu_{B2^i} \subset k^\times$ be the group of $B2^i$-th roots of unity, so $|D_i| = B2^i$ and $x \mapsto x^2$ maps D_i two-to-one onto D_{i-1}. The shrinking domains are $E_j := D_{m-j}$. The fibers of the top domain over D_0 are

$$\Phi_z := \{y \in D_m : \ y^L = z\}, \qquad z \in D_0, \qquad |\Phi_z| = L.$$

Index conventions. For $S \subseteq [m]$ put

$$j(S) := \sum_{k \in S} 2^{k-1}, \qquad j'(S) := \sum_{k \in S} 2^{m-k}.$$

Both $S \mapsto j(S)$ and $S \mapsto j'(S)$ are bijections $2^{[m]} \to \{0, \ldots, L-1\}$. For an integer $0 \leq t < L$, t_k denotes the bit of weight 2^{k-1}, so $t = \sum_k t_k 2^{k-1}$. For $1 \leq i \leq m$ and $0 \leq t < 2^i$ let $\mathrm{rev}_i(t) := \sum_{k=1}^{i} t_k 2^{i-k}$.

Polynomials. The multilinear $f = \sum_{S \subseteq [m]} c_S X^S$ (with $X^S = \prod_{k \in S} X_k$) has Kronecker encoding

$$F(Y) := f(Y, Y^2, \ldots, Y^{2^{m-1}}) = \sum_S c_S Y^{j(S)}, \qquad \deg F < L.$$

This is a K-linear bijection between multilinear polynomials and $K[Y]_{<L}$; we write $f = \kappa^{-1}(F)$. For $r = (r_1, \ldots, r_m) \in K^m$ define

$$\beta_k(X) := r_k X^{2^{m-k}}, \qquad f(\beta(X)) := f(\beta_1(X), \ldots, \beta_m(X)) = \sum_S c_S \Big(\prod_{k \in S} r_k \Big) X^{j'(S)}.$$

For $0 \leq i \leq m$ let $A_i(X_1, \ldots, X_i) := \sum_{b \in \{0,1\}^{m-i}} f(X_1, \ldots, X_i, b)$, so $A_m = f$ and $A_0 = \sum_b f(b)$.

Even/odd split. Every $P \in K[X]$ is uniquely $P(X) = E_P(X^2) + X\, O_P(X^2)$.

Codes. $\mathrm{RS} := \mathrm{RS}[K, D_m, L] = \{F|_{D_m} : F \in K[Y]_{<L}\}$. Relative Hamming distance is $\mathrm{d}(\cdot,\cdot)$. The minimum relative distance of RS is $\lambda = 1 - \rho + 1/n > 2\theta$.

Remark 1.1 (The target is an evaluation)**.** Since $\sum_b \prod_{k\in S} b_k = 2^{m-|S|}$,

$$\sum_{b\in\{0,1\}^m} f(b) = \sum_S c_S\, 2^{m-|S|} = 2^m f\big(\tfrac{1}{2},\dots,\tfrac{1}{2}\big).$$

A hypercube sum of a single committed multilinear polynomial is therefore an evaluation claim at the point $(\frac{1}{2},\dots,\frac{1}{2})$. This identity explains the structure found below: both chains are re-encodings of the coefficient vector $(c_S)_S$, and the sum is a fixed linear functional of it.

2 The protocol

2.1 Chain A: growing-domain Kronecker–Sumcheck

Oracles: $P_0 := H$ (a scalar); for $1 \le i \le m$, tables $U_i, V_i : D_{i-1} \to K$ and $P_i : D_i \to K$.

Round i: the prover commits to (U_i, V_i); the verifier samples r_i; the prover commits to P_i. The intended relations are

$$P_{i-1}(Y) = 2U_i(Y) + V_i(Y), \qquad P_i(X) = U_i(X^2) + r_i X\, V_i(X^2).$$

Path check. For a point $x_m \in D_m$ let $x_{i-1} := x_i^2$. The verifier opens, for $i = 1,\dots,m$:

$$\begin{aligned} &(\mathrm{a}_i) \quad P_{i-1}(x_{i-1}) \stackrel{?}{=} 2U_i(x_{i-1}) + V_i(x_{i-1}) \qquad (P_0 := H),\\ &(\mathrm{b}_i) \quad P_i(x_i) \stackrel{?}{=} U_i(x_{i-1}) + r_i x_i\, V_i(x_{i-1}). \end{aligned}$$

2.2 Chain B: shrinking-domain polynomial-valued FRI

Leaf format. A level-i leaf ($0 \le i \le m$) sits at $z \in E_i = D_{m-i}$ and is an element

$$s_i(z,\cdot) \in K[W]_{<2^i}, \qquad \text{representing } S_i(z,X) := s_i(z, X^{2^{m-i}}).$$

It is committed as its canonical vector of 2^i coefficients. Level 0 is the committed table $s_0(y) \in K$, $y \in D_m$ (the oracle C_f).

Fold. For $1 \le i \le m$, $y \in E_{i-1}$, and $a, b \in K[W]_{<2^{i-1}}$ put $e := \frac{a+b}{2}$, $o := \frac{a-b}{2y}$ and

$$\mathrm{Fold}_{i,y}(a,b)(V) := e(V^2) + r_i V\, o(V^2) \ \in K[V]_{<2^i}.$$

With $V = X^{2^{m-i}}$ and $W = V^2 = X^{2^{m-i+1}}$, this is exactly

$$S_i(y^2, X) = \frac{S_{i-1}(y,X) + S_{i-1}(-y,X)}{2} + r_i X^{2^{m-i}}\, \frac{S_{i-1}(y,X) - S_{i-1}(-y,X)}{2y}.$$

In round i, after r_i, the prover commits to the level-i table $z \mapsto s_i(z,\cdot)$ on E_i.

Lemma 2.1 (Injectivity of the fold)**.** *If $r_i \neq 0$ and $y \neq 0$, then $\mathrm{Fold}_{i,y} : K[W]^2_{<2^{i-1}} \to K[V]_{<2^i}$ is a linear bijection.*

Proof. $e(V^2)$ is supported on even powers and $r_i V o(V^2)$ on odd powers, so the image determines (e, o); then $a = e + yo$, $b = e - yo$. Dimensions agree ($2 \cdot 2^{i-1} = 2^i$). □

Chain-B query. For $y_0 \in D_m$ let $y_i := y_{i-1}^2 \in E_i$. The verifier opens $s_i(\pm y_i)$ and $s_{i+1}(y_{i+1})$ for $0 \le i < m$ (pairs $\{y, -y\}$ are grouped in one Merkle leaf), checks the leaf format, and checks $s_{i+1}(y_{i+1}) = \mathrm{Fold}_{i+1,y_i}\big(s_i(y_i), s_i(-y_i)\big)$.

2.3 The repaired protocol

The three items marked (F1)–(F3) are not optional; Section 5 shows what fails without (F1) and why (F2) is load-bearing.

Protocol 2.2 (Π_{dual})**.**

1. Prover sends C_f (oracle $s_0 : D_m \to K$) and H.
2. For $i = 1, \dots, m$: prover commits (U_i, V_i) on D_{i-1}; verifier sends $r_i \leftarrow K$; prover commits P_i on D_i and the chain-B level s_i on E_i.
3. Query phase (all commitments fixed):
   - (F1) Open all B terminal leaves $s_m(z, \cdot)$, $z \in D_0$, in full; reject unless they are equal. Call the common value $T \in K[X]_{<L}$.
   - (F2) Every opened leaf must be a coefficient vector of length exactly 2^i at level i.
   - (F3) Run q chain-B queries from independent uniform $y_0^{(1)}, \dots, y_0^{(q)} \in D_m$.
   - (A) Run one chain-A path check from a point $x_m \in D_m$ (any choice, e.g. $x_m = y_0^{(1)}$), and check $P_m(x_m) = T(x_m)$.
4. Accept iff all checks pass.

3 Completeness

Lemma 3.1 (Honest chain A)**.** *Define, for $1 \le i \le m$,*

$$Q_i(Y, Z) := A_i\big(r_1 Y^{2^{i-2}}, r_2 Y^{2^{i-3}}, \dots, r_{i-1} Y,\ Z\big), \qquad Q_1(Y, Z) := A_1(Z),$$

write $Q_i = U_i(Y) + Z\, V_i(Y)$, and set $P_i(X) := A_i(r_1 X^{2^{i-1}}, \dots, r_{i-1} X^2, r_i X)$, $P_0 = A_0$. Then for all i:

(i) Q_i is affine in Z, and $\deg U_i, \deg V_i \le 2^{i-1} - 1$;

(ii) $P_{i-1}(Y) = 2U_i(Y) + V_i(Y)$;

(iii) $P_i(X) = U_i(X^2) + r_i X\, V_i(X^2)$;

(iv) $P_m(X) = f(\beta(X))$.

Proof. (i) A_i is multilinear; the Y-degree is at most $\sum_{t=0}^{i-2} 2^t = 2^{i-1} - 1$. (ii) $Q_i(Y, 0) + Q_i(Y, 1) = A_{i-1}(r_1 Y^{2^{i-2}}, \dots, r_{i-1} Y) = P_{i-1}(Y)$, using $A_{i-1}(\cdot) = A_i(\cdot, 0) + A_i(\cdot, 1)$. (iii) $Q_i(X^2, r_i X) = A_i(r_1 X^{2^{i-1}}, \dots, r_{i-1} X^2, r_i X) = P_i(X)$. (iv) is (iii)'s definition at $i = m$, since $A_m = f$. □

Lemma 3.2 (Honest chain B)**.** *Let $F = \sum_{t<L} c_t Y^t$ and $s_0 = F|_{D_m}$, and fold honestly. Then for $0 \le i \le m$ and $z \in E_i$,*

$$s_i(z, V) = \sum_{t<L} c_t\ z^{\lfloor t/2^i \rfloor} \Big(\prod_{k \le i,\ t_k = 1} r_k \Big)\, V^{\mathrm{rev}_i(t \bmod 2^i)}.$$

In particular $\deg_V s_i < 2^i$, and $S_m(z, X) = s_m(z, X) = f(\beta(X))$ for every $z \in D_0$.

Proof. Induction on i; $i = 0$ is the definition. Assume the formula at level $i - 1$ and fix $y \in E_{i-1}$, $z = y^2$. Write $u = \lfloor t/2^{i-1} \rfloor$, so $t_i = u \bmod 2$. Then $\frac{y^u + (-y)^u}{2} = y^u[t_i = 0]$ and $\frac{y^u - (-y)^u}{2y} = y^{u-1}[t_i = 1]$, and in both cases $y^{u - t_i} = z^{\lfloor t/2^i \rfloor}$. The term with $t_i = 0$ goes to $V^{2\mathrm{rev}_{i-1}(t)}$, the term with $t_i = 1$ to $r_i V^{1 + 2\mathrm{rev}_{i-1}(t)}$, and $\mathrm{rev}_i(t \bmod 2^i) = 2\mathrm{rev}_{i-1}(t \bmod 2^{i-1}) + t_i$. At $i = m$: $\lfloor t/L \rfloor = 0$ and $X^{\mathrm{rev}_m(j(S))} = X^{j'(S)}$. □

Corollary 3.3 (Completeness)**.** *The honest prover is accepted with probability* 1*: all leaves are well formed, all B terminal leaves equal $f(\beta) = P_m$, and every local check is an instance of Lemma 3.1 or 3.2.*

Remark 3.4 (An earlier indexing bug)**.** With the tempting definition

$$Q_i(Y, Z) = A_i\big(r_1 Y^{2^{i-1}}, \dots, r_{i-1} Y^2, Z\big),$$

identity (ii) would require the first of the following, whereas Lemma 3.1 gives the second:

$$P_{i-1}(Y) \stackrel{?}{=} A_{i-1}\big(r_1 Y^{2^{i-1}}, \dots, r_{i-1} Y^2\big), \qquad P_{i-1}(Y) = A_{i-1}\big(r_1 Y^{2^{i-2}}, \dots, r_{i-1} Y\big);$$

the two differ for $i \geq 2$ in general. Lemma 3.1 is the consistent orientation.

4 The exact-identity model

Suppose every round relation holds as a polynomial identity, all $r_i \neq 0$, and the leaves have the format (F2).

Proposition 4.1 (Chain A is invertible)**.** *The map $(U_i, V_i) \mapsto P_i$ with $P_{i-1} = 2U_i + V_i$ is inverted by $U_i = E_{P_i}$, $V_i = r_i^{-1} O_{P_i}$, so $P_{i-1} = 2E_{P_i} + r_i^{-1} O_{P_i}$. Iterating,*

$$H = P_0 = \Lambda_r(P_m), \qquad \Lambda_r\Big(\sum_S p_S\, X^{j'(S)}\Big) := \sum_S p_S\, 2^{m-|S|} \prod_{k \in S} r_k^{-1}.$$

If $P_m = f(\beta)$, i.e. $p_S = c_S \prod_{k \in S} r_k$, then $H = \sum_S c_S 2^{m-|S|} = \sum_b f(b)$.

Proof. At round i the lowest power X^1 of P_i carries variable X_i, so the odd part corresponds to $i \in S$; the inverse step multiplies coefficients with $i \notin S$ by 2 and those with $i \in S$ by r_i^{-1}. Compose over $i = m, \dots, 1$ and use Remark 1.1. □

Proposition 4.2 (Chain B is a bijective re-encoding)**.** *For fixed r with all $r_k \neq 0$, $f \mapsto f(\beta)$ is a bijection from multilinear polynomials onto $K[X]_{<L}$, and by Lemma 2.1 each fiber Φ_z of level* 0 *is mapped bijectively onto the terminal leaf $s_m(z, \cdot)$.*

Remark 4.3. In the exact-identity model, neither chain uses the randomness of r: chain A computes the fixed functional Λ_r, which on honest inputs is the r-independent value $\sum_b f(b)$, and chain B is a deterministic butterfly of F. In particular a prover cannot satisfy the recurrences with an alternative polynomial family, since the recurrences are invertible. All soundness content therefore lies in the passage from sampled checks to identities, which is where the query analysis of Section 6 lives.

5 Attacks on under-specified variants

5.1 Fiber splitting: false sum accepted with probability one

Let Π^- be Π_{dual} with (F1) replaced by: *every check that needs a terminal value at x_m (resp. at the end of a chain-B query) uses the leaf $s_m(x_m^L, \cdot)$ (resp. $s_m(y_m, \cdot)$) of its own fiber.*

Observe that both chains split into B independent subtrees indexed by $z \in D_0$: in chain B, $\pm y$ and y^2 lie over the same z; in chain A, x and x^2 satisfy $x^{2^i} = (x^2)^{2^{i-1}}$.

Proposition 5.1. *For every multilinear $f_\star$ and every $H' \neq \sum_b f_\star(b)$ there is a prover for Π^- whose initial table satisfies $\mathrm{d}(s_0, F_\star) = B/n = 1/L$ (so $F_\star = \kappa(f_\star)$ is its unique decoding) and which is accepted with probability* 1 *claiming H'.*

Proof. For $z \in D_0$ pick $y_z \in \Phi_z$ and set

$$h_z(Y) := \frac{Y^L - z}{Y - y_z} = \sum_{t=0}^{L-1} y_z^{L-1-t} Y^t.$$

It vanishes on $\Phi_z \setminus \{y_z\}$ and $h_z(y_z) = L y_z^{L-1} \neq 0$. Let $\eta_z := \kappa^{-1}(h_z)$. By Remark 1.1,

$$\sigma_z := \sum_b \eta_z(b) = \sum_{t<L} y_z^{L-1-t} 2^{m-\mathrm{wt}(t)} = y_z^{L-1} \prod_{k=1}^{m} \left(2 + y_z^{-2^{k-1}}\right).$$

For each k at most 2^{k-1} elements of the coset Φ_z satisfy $y^{2^{k-1}} = -\frac{1}{2}$, so at most $L - 1 < |\Phi_z|$ choices are excluded, and we may choose y_z with $\sigma_z \neq 0$. Put $\lambda_z := (H' - H_\star)/\sigma_z$ with $H_\star := \sum_b f_\star(b)$, and

$$s_0(y) := F_\star(y) + \lambda_z h_z(y) \qquad (y \in \Phi_z).$$

This differs from $F_\star$ exactly at the B points y_z. On subtree z the prover runs the honest prover (both chains, Lemmas 3.1 and 3.2) for $g_z := f_\star + \lambda_z \eta_z$, whose Kronecker encoding interpolates s_0 on Φ_z. Every local check holds identically, the leaf $s_m(z, \cdot) = g_z(\beta)$ matches chain A on the same subtree, and $\sum_b g_z(b) = H_\star + \lambda_z \sigma_z = H'$ for every z, consistent with the single scalar $P_0 = H'$. □

Under (F1) this prover is rejected with certainty: $g_z(\beta) = g_{z'}(\beta)$ would force $\lambda_z h_z = \lambda_{z'} h_{z'}$, but h_z vanishes on $\Phi_z \setminus \{y_z\}$ while $h_{z'}(y) = (z - z')/(y - y_{z'}) \neq 0$ there.

5.2 Late commitment of the first round message

Proposition 5.2. *If (U_1, V_1) is committed after r_1, then for any $H' \neq H_\star$ there is a prover that is honest in chain B and passes each independent uniform chain-A path with probability $1/2$.*

Proof. For each $y \in D_0$ choose a root $x_y \in D_1$ with $2r_1 x_y \neq 1$ and solve $2U + V = H'$, $U + r_1 x_y V = P_1^\star(x_y)$ (determinant $2r_1 x_y - 1$). Commit $P_1 := P_1^\star$ and play honestly from round 2 on. Only (b_1) can fail, and only when $x_1 = -x_y$; for uniform x_m, $x_1 = x_m^{2^{m-1}}$ is uniform on D_1. □

With the correct order the corresponding error is at most $2B(2^m - 1)/|K|$, about 2^{-169} (Lemma 6.3), so the late variant drops chain A from field-size soundness to 2^{-q_A} for q_A paths.

5.3 Why (F2) is load-bearing

The detection argument (Lemma 6.5) walks up a query path and uses injectivity of $\mathrm{Fold}_{k,y}$ on the opened pair. On a query path only the path node is checked against its own children; the sibling is not. Lemma 2.1 gives injectivity only on $K[W]^2_{<2^{k-1}}$, and without the format check a sibling could be an arbitrary polynomial of larger degree, for which $(a, b) \mapsto e(V^2) + r_i V o(V^2)$ has a nontrivial kernel. (F2) costs nothing and also fixes the leaf size at 2^i coefficients, which is what makes the cost bounds of Section 7 hold.

6 Soundness

6.1 The binding game

Definition 6.1 (Reference polynomial)**.** Given the committed oracle $s_0 : D_m \to K$, let $F_\star$ be the unique $F \in K[Y]_{<L}$ with $\mathrm{d}(s_0, F|_{D_m}) < \theta$, if it exists (unique because $2\theta < \lambda$), and $f_\star := \kappa^{-1}(F_\star)$, $H_\star := \sum_b f_\star(b)$.

Definition 6.2 (Game)**.** An unbounded interactive oracle prover $\widetilde{P}$ interacts with the verifier of Protocol 2.2. $\widetilde{P}$ wins if the verifier accepts and either $F_\star$ does not exist or $H \neq H_\star$.

Note that $F_\star$ is a function of s_0 alone, hence fixed before r_1 is sampled.

6.2 Chain A: pathwise sumcheck

Let $U_i^\star, V_i^\star, P_i^\star$ be the honest tables of Lemma 3.1 for $f_\star$; they depend on $f_\star$ and $r_1, \dots, r_{i-1}$ only (and $P_i^\star$ also on r_i).

Lemma 6.3 (Pathwise sumcheck)**.** *Assume $F_\star$ exists. Let Bad_A be the event that for some i, some $y \in D_{i-1}$ and some x with $x^2 = y$,*

$$(U_i(y), V_i(y)) \neq (U_i^\star(y), V_i^\star(y)) \quad \textit{and} \quad U_i(y) + r_i x V_i(y) = U_i^\star(y) + r_i x V_i^\star(y).$$

Then $\Pr[\mathsf{Bad}_A] \leq \sum_{i=1}^{m} 2|D_{i-1}|/|K| = 2B(2^m - 1)/|K|$. *Moreover, outside Bad_A, for* every $x_m \in D_m$*: if $H \neq H_\star$ and all checks $(\mathrm{a}_i), (\mathrm{b}_i)$ on the path from x_m pass, then $P_m(x_m) \neq f_\star(\beta(x_m))$.*

Proof. Conditioned on the transcript before r_i, the tables $U_i, V_i, U_i^\star, V_i^\star$ are fixed and r_i is uniform. For fixed (y, x) with $(\Delta U, \Delta V) := (U_i - U_i^\star, V_i - V_i^\star)(y) \neq 0$ and $x \neq 0$, the affine function $r \mapsto \Delta U + rx\Delta V$ has at most one root. A union bound over i, y and the two roots x gives the probability bound.

For the second claim let $\Delta_i := P_i(x_i) - P_i^\star(x_i)$, with $\Delta_0 = H - H_\star \neq 0$. If $\Delta_{i-1} \neq 0$ and (a_i) passes, then $2\Delta U + \Delta V = \Delta_{i-1} \neq 0$ at x_{i-1}, so $(\Delta U, \Delta V) \neq 0$; outside Bad_A the right-hand side of (b_i) differs from $P_i^\star(x_i)$, and since (b_i) passes, $\Delta_i \neq 0$. By induction $\Delta_m \neq 0$, and $P_m^\star = f_\star(\beta)$ by Lemma 3.1(iv). □

Remark 6.4. Lemma 6.3 needs no degree test on P_i, U_i, V_i, and one path suffices for any choice of x_m, including a fixed one. Wrongness propagates under pullback: a wrong value at y forces wrong values at both square roots, so the error pattern of a false claim covers all of D_m, the opposite of FRI's distance decay.

6.3 Chain B: proximity-gap-free tree detection

Lemma 6.5 (Tree detection)**.** *Assume all $r_k \neq 0$, (F1) and (F2) pass with common terminal value T, and let $F' \in K[Y]_{<L}$ be the unique polynomial with $\kappa^{-1}(F')(\beta) = T$ (Proposition 4.2). Then every chain-B query started at y_0 with $s_0(y_0) \neq F'(y_0)$ rejects. Consequently*

$$\Pr\left[\textit{all } q \textit{ chain-B queries accept} \;\middle|\; \textit{commitments}\right] \leq \left(1 - \mathrm{d}(s_0, F')\right)^q.$$

Proof. Let s_i' be the honest butterfly of F' (Lemma 3.2); then $s_m'(z) = T$ for all $z \in D_0$, so $W_m = \emptyset$ where $W_i := \{z \in E_i : s_i(z) \neq s_i'(z)\}$. Let $y_0 \in W_0$ with ancestors $y_1, \dots, y_m$, and let $k \geq 1$ be minimal with $y_k \notin W_k$. If the check at y_k passed, then

$$\mathrm{Fold}_{k,y_{k-1}}\big(s_{k-1}(y_{k-1}), s_{k-1}(-y_{k-1})\big) = s_k(y_k) = s_k'(y_k) = \mathrm{Fold}_{k,y_{k-1}}\big(s_{k-1}'(y_{k-1}), s_{k-1}'(-y_{k-1})\big),$$

and since the opened leaves have the format (F2), Lemma 2.1 gives $s_{k-1}(y_{k-1}) = s_{k-1}'(y_{k-1})$, contradicting $y_{k-1} \in W_{k-1}$. The queries are uniform and independent after T, hence F', is fixed. □

Lemma 6.6 (Distance to a wrong target)**.** *If $F_\star$ does not exist, or exists and $F' \neq F_\star$, then* $\mathrm{d}(s_0, F') \geq \theta$.

Proof. If $F_\star$ does not exist, $\mathrm{d}(s_0, c) \geq \theta$ for every codeword c. Otherwise $\mathrm{d}(s_0, F') \geq \lambda - \mathrm{d}(s_0, F_\star) > \lambda - \theta > \theta$. □

6.4 Main theorem

Theorem 6.7 (Interactive soundness of Π_{dual})**.** *For every unbounded interactive oracle prover,*

$$\Pr[\mathsf{Win}] \;\le\; \frac{m}{|K|} + \frac{2B(2^m-1)}{|K|} + (1-\theta)^q.$$

Proof. Partition the winning event.

(1) Some $r_k = 0$: probability at most $m/|K|$.

(2) All $r_k \neq 0$, and $F_\star$ does not exist or $T \neq f_\star(\beta)$. Then F' is well defined and either $F_\star$ is absent or $F' \neq F_\star$, so by Lemmas 6.5 and 6.6 acceptance has probability at most $(1-\theta)^q$.

(3) All $r_k \neq 0$, $F_\star$ exists, $T = f_\star(\beta)$, and $H \neq H_\star$. The check $P_m(x_m) = T(x_m) = f_\star(\beta(x_m))$ together with the chain-A path checks contradicts Lemma 6.3 unless Bad_A holds. □

6.5 Parameters

For $K = \mathbb{F}_{p^3}$ ($|K| \approx 2^{192}$), $m = 20$, $B = 4$ ($\theta = 3/8$):

Term	Value	$\log_2$
$m/\|K\|$	$20/p^3$	-187.7
$2B(2^m-1)/\|K\|$	$8(2^{20}-1)/p^3$	-169.0
$(5/8)^{189}$		-128.16
$(5/8)^{188}$		-127.48

Hence $q = 189$ chain-B queries and a single chain-A path give total interactive error below 2^{-128}, and $q = 189$ is minimal for this bound. This coincides with the unique-decoding query count of the line-binding symbolic-FRI protocol, as expected: both are governed by $(1-\theta)^q$.

Remark 6.8 (Compilation)**.** Under the BCS transform with Merkle commitments in the random-oracle model, add $\varepsilon_{\mathrm{Merkle}}$, the probability that the prover finds a collision (at most $Q^2/2^{\lambda_h}$ for Q queries and λ_h-bit digests). For the Fiat–Shamir NIZK, the query phase is a single round with error $(1-\theta)^q$ per attempt, so a prover with Q hash queries gains a factor of about Q; compensate with grinding or with $q \ge (128 + \log_2 Q - g)/\log_2(8/5)$ for g bits of proof-of-work.

Remark 6.9 (Beyond unique decoding)**.** Lemma 6.5 gives per-query detection $\mathrm{d}(s_0, F')$ for the target F' chosen by the prover. Above θ the target may be any codeword of the list around s_0, selected adaptively after r, so the winning condition must change to a list statement; the deterministic chain alone cannot bind to a single $f_\star$ there.

7 Costs

All counts are in elements of K (24 bytes); $m = 20$, $B = 4$, $q = 189$.

Prover.

- Chain B: level i has $B2^{m-i}$ leaves of 2^i coefficients, i.e. BL elements per level, $mBL \approx 8.39 \cdot 10^7$ elements (≈ 2.0 GB) in total, computed in $O(mBL)$ field operations.
- Chain A: $\sum_{i=1}^{m} 2B2^i \approx 4BL \approx 1.68 \cdot 10^7$ elements; the coefficients of all A_i come from $O(L)$ partial sums, then one NTT per table.

Proof. At level $i < m$ the opened data is at most $\min(q, B2^{m-i-1}) \cdot 2^{i+1}$ elements.

Levels	Opened elements (upper bound)
0–13	$189\,(2^{15} - 2) \approx 6.19 \cdot 10^6$
14–19 (whole levels)	$6 \cdot BL \approx 2.52 \cdot 10^7$
terminal leaves (F1)	$BL \approx 4.19 \cdot 10^6$
chain A (one path)	$3m = 60$
total	$\approx 3.56 \cdot 10^7$ (≈ 0.85 GB)

Verifier. $\Theta(L)$: reading T, evaluating $T(x_m)$, and fold checks whose cost equals the opened data. Once T is read, the verifier could compute $H = \Lambda_r(T)$ directly, so in Π_{dual} chain A is redundant.

8 Collapse: specializing the formal variable

Lemma 8.1 (Specialization)**.** *Fix $x \in k^\times$ and put $\alpha_i := r_i x^{2^{m-i}}$. Then:*

(i) the scalars $g_i(z) := S_i(z, x)$ satisfy the scalar FRI fold

$$g_i(y^2) = \frac{g_{i-1}(y) + g_{i-1}(-y)}{2} + \alpha_i \frac{g_{i-1}(y) - g_{i-1}(-y)}{2y};$$

(ii) along the chain-A path from $x_m = x$, with $(u_i, v_i) := (U_i, V_i)(x_{i-1})$, the checks $(\mathrm{a}_i), (\mathrm{b}_i)$ read

$$P_{i-1}(x_{i-1}) = 2u_i + v_i, \qquad P_i(x_i) = u_i + \alpha_i v_i,$$

which is classical sumcheck with round polynomial $u_i + Zv_i$ and challenge α_i;

(iii) the α_i are independent and uniform in K.

Proof. (i) and (ii) are substitution of $X = x$; note $x_i = x^{2^{m-i}}$ so $r_i x_i = \alpha_i$. (iii) $r_i \mapsto r_i x^{2^{m-i}}$ is a bijection of K. □

The weighting $X^{2^{m-i}}$ is therefore cosmetic once X is specialized, and the same α_i drives both chains. Taking $x = 1$:

Protocol 8.2 (Π_1)**.** Commit $s_0 = F|_{D_m}$ and send H. For $i = 1, \ldots, m$: send $(u_i, v_i) \in K^2$ in the clear, check $2u_i + v_i = c_{i-1}$ ($c_0 = H$), sample r_i, set $c_i = u_i + r_i v_i$, commit the scalar FRI layer g_i on D_{m-i} folded with r_i. Finally check that g_m is constant on D_0 with value c_m, and run q FRI queries.

Π_1 is classical sumcheck for $\sum_b f(b)$ interleaved with scalar FRI on the Kronecker encoding, using shared challenges; this is the BaseFold pattern [5] in the monomial basis, and it is the scalar-reduction structure of Sum-FRI.

Theorem 8.3 (Soundness of Π_1, conditional)**.** *Assume the evaluation-binding property of scalar FRI in the unique-decoding regime: for every prover, $\Pr[\textit{FRI accepts} \wedge (F_\star \textit{ absent} \vee c \neq f_\star(r))] \leq \varepsilon_{\text{bind}}(q)$, where c is the common final value. Then*

$$\Pr[\mathsf{Win}] \leq \frac{m}{|K|} + \varepsilon_{\text{bind}}(q).$$

Proof. Outside the binding event, $c_m = f_\star(r)$, and Lemma 6.3 with the single point $x = 1$ (one pair $(y, x) = (1, 1)$ per round) bounds the remaining event $H \neq H_\star$ by $m/|K|$. Both bad events are defined relative to the same $f_\star$, which is fixed before r_1, so the union bound is valid despite the shared challenges. □

The hypothesis of Theorem 8.3 is precisely where a proximity-gap or correlated-agreement theorem enters: the scalar fold is not injective, so Lemma 6.5 does not transfer. In the unique-decoding regime it is supplied by the BaseFold analysis [5]; in the list-decoding regime see [6]. The polynomial-valued chain of Π_{dual} buys exactly the removal of this hypothesis, at the price of a $\Theta(L)$ proof.

Cost of Π_1. $2m$ scalars of sumcheck, and per query one pair and one Merkle path per level. With pairs grouped in leaves and 32-byte digests this is $\sum_{i=0}^{m-1}\big(48 + 32(m-i+1)\big) = 8320$ bytes per query, about 1.57 MB for $q = 189$ before path deduplication.

9 Summary of the security relation

	Π_{dual} (formal X)	Π_1 ($X := 1$)
Chain A	pathwise sumcheck, one path, $\leq 2^{-169}$	classical sumcheck, $m/\lvert K\rvert$
Chain B binding	injective fold, deterministic, $(1-\theta)^q$	scalar FRI, needs a proximity-gap theorem
Role of r in chain B	none	essential
Terminal link	B full leaves equal, then $P_m(x_m) = T(x_m)$	final layer constant $= c_m$
Proof size ($m = 20$)	≈ 0.85 GB	≈ 1.6 MB
Verifier	$\Theta(L)$	$O(q\,m)$

The dual-chain design thus separates cleanly into a part that needs no proximity theory but is not succinct (the formal butterfly) and a succinct part whose binding is exactly the evaluation-binding problem for scalar FRI. Remark 1.1 shows that, for a single multilinear polynomial, the sum is the evaluation $2^m f(\frac{1}{2}, \ldots, \frac{1}{2})$; the sumcheck layer becomes necessary once the summand is a product (for instance $f \cdot g$ or $f \cdot \mathrm{eq}$), which is the natural next target for this construction.

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